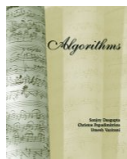


CS 170:Algorithm



Prof Elchanan Mossel.

Based on: Algorithms by Dasgupta Papadimitriou and Vazirani.

Slides edited from a version created by Prof. Satish Rao.

For UC-Berkeley CS170 Spring 2014 students use only.

Do not re-post or distribute

CS 170: Algorithms

Static Course Webpage. (inst.cs.berkeley.edu/cs170)

Will mostly use *Piazza*. Should be signed up. Otherwise, sign up on your own.

- ▶ Homework will be turned in electronically.
- ▶ Latex ok, formatting ok, scanning handwritten stuff ok.
- ▶ If scan, compress to at most 3 MB in size.
- ▶ Instructions on web-page.

Don't cheat! HW are just 0.1 of grade. **All suspected cheaters will be reported to office of students conduct.**

Always write all of your sources including friends you are working with and online resources.

Instructor.

Prof. Elchanan Mossel

Statistics & Computer Science

School: PhD in math, long ago, far away

... Worked as algorithm engineer for 5 years

... After PhD, Microsoft Research

... Berkeley!

(also consulted at Microsoft, social networks, bio startups;
many years ago ... a popular babysitter)

GSIs

Rudy Dai

Berkeley Undergrad

Lewin Gan

Berkeley Undergrad

Edwin Liao

Berkeley Undergrad

Yun Park

Berkeley Undergrad

Atif Rahman

PhD student

Bioinformatics

Ning Tan

PhD student

Theoretical Computer Science

Ben Weitz

PhD Student

Theoretical Computer Science

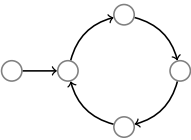
Puzzler.

Does a list have a cycle?

Algorithm: Mark first node, see if one comes back to it.

Pep Quiz: Does this work?

a) Yes. b) No. c) Not sure.



First node not in cycle!

Answer is no.

Does a list have a cycle?

Two pointers p1, p2:

advance p1 twice

advance p2 once.

If ever at the same place, report cycle.

Stop when ptr 1 is stuck.

Pet Quiz: Does this work?

a) Yes. b) No. c) Not sure.

Does a list have a cycle?

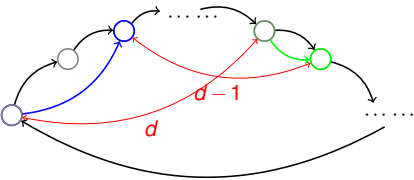
Correctness:

If no cycle, slow pointer never catches fast one.

If cycle, both pointers will enter cycle at some time.

d - distance from fast ptr to slow ptr.

d decreases every step.



Runtime: n steps to cycle

n steps to catch up.

$O(n)$

Additional storage: two pointers. $O(1)$.

Puzzles & Algorithm

Solutions to puzzles
..are Algorithms...
if they
..are correct...and efficient.
Is this a useful process?

Algorithms for the Human Genome Project

Reconstruct DNA...
ACTGAAACTGAGTAGATA....
Read first, then next, then next, ...3.1 billion times...
.. slow... error prone...
Parallel sequencing yields chunks of overlapping DNA.
AGTAG, AGATA, TGAGT , ACTGAA , CTGAA , AAACTG
Assemble into a consistent string?
ACTGAA
 AAACTG
 TGAGT
 AGTAG
 AGATA

ACTGAAACTGAGTAGATA
Correctness and efficiency still studies extensively (e.g.
cancers ...)

Page Rank.

Problem: What is good on the web?
Website that pays search engine most? Goto.com.

[Random Surfer Model \(Brin-Page\)](#): Follow link, follow link,
.. occasionally jump to random page (with prob. ϵ).

Popular pages are desirable pages.
PageRank = popularity for random surfer.
Sort search results by PageRank!
Made us happy.
Google.
Correctness and efficiency studied extensively
E.g: If a tree falls in the forest ...

Algorithms...

Driving Directions
Airline Scheduling
Compiling
Compression
Cryptography
Optimization
.
.
.

Doing well in this class..

The Bloom 2-sigma effect.

One on one instruction and mastery learning leads to a two sigma improvement in performance.

50th percentile to the 98th percentile.

The median Harvard applicant would get in!

One on one instructionfor 450 students ???

Resources: office hours, discussion, piazza

and .. you! and your friends!

You have the book, home-works, exams, slides, and the web...

The book.

Close reading.

Work through examples.

Take care to understand arguments.

Homeworks.

Do them, in groups (up to four ; write names of group members).

Write up solutions, on your own, in your own words.

Uh...helpful for exams.

From Middle Ages to Renaissance

Two big revolutions around the same time:

Print (dissemination of information)

Decimal system (algorithms for organizing information)

Huge impact on commerce, astronomy, geography, math

Calculating: 300 BC through Middle Ages in Europe.

I – one

V – five

X – ten

L – fifty

C – one hundred

D – five hundred

M – a thousand

VIII – eight

DCLXV – five hundred plus a hundred plus fifty plus ten plus five

MCDLXVIII – one thousand five hundred minus one hundred

Add them?

$1448 + 666 = 2014$

666 years since the printing press.

Multiply roman numbers?

Modern system.

From India, via Al Khwarizmi.

He also described recipes for adding, multiplying, computing digits of π ..

Algorithms!

Note:

Mayans (base 20): dots (ones) and underlines (fives).

13 is “..”

Babylonians (base 60): clusters of 10 instead of digits.

India: “invented” 0! ... and decimal symbols.

20th century. Base 2!

The input representation for modern computers and communication.

Writing to propagating..

Al Khwarizmi: Go west! Young decimal system!

Al Khwarizmi used to be transliterated as *Algoritmi* or Algaurizin

Persian mathematician, astronomer, geographer (780-850)

..but Fibonacci popularized its use.

So some homage...

Italian mathematician (1170-1250) who traveled to learn the Hindu-Arab math.



Fibonacci numbers.

Growth of population of rabbits (why ???)

$F_0 = 0, F_1 = 1.$

$F_n = F_{n-1} + F_{n-2}.$

```
def fib(n):  
    if n <= 1:  
        return n  
    else:  
        return fib(n-1) + fib(n-2)
```

Correct? Implements definition!

Run time.

$T(n) = T(n-1) + T(n-2) + 2$

$T(n) \geq F_n$

Fibonacci algorithm and numbers!

$$F_n = F_{n-1} + F_{n-2} = F_{n-2} + F_{n-3} + F_{n-2} \geq 2F_{n-2}$$

By induction, we get $F_n \geq 2^{n/2}$.

From book.. $F_n \approx 2^{0.694n}$.

$$T(n) \geq 2^{n/2}$$

From book $T(n) \geq 2^{0.694n}$

For $n = 100$, this is around 2^{64} operations, (much more than thousand years or so on a fast computer.)

Exponential algorithm. Bad. Grows very fast.

Can we do better?

Sure..much better.

```
def fib(n):
    if n <= 1:
        return n
    else:
        a = [0,1]
        for i in xrange(2,n+1):
            a.append(a[i-1]+a[i-2])
        return a[n]
```

$O(n)$ operations! Maybe.

Size matters.

How many bits in F_n ?

Remember $F_n \approx 2^{0.694n}$.

About how many bits in F_n ?

$$\log_2 F_n \approx 0.6294n$$

How long does it take to compute $F_{n-1} + F_{n-2}$?

$O(n)$.

How long does Fib take?

n additions.

At most $O(n^2)$.

Polynomial.

Doubling size, made time grow by factor of four.

cn^2 runtime.

$$c(2n)^2 = 4cn^2.$$

Asymptotic bounds give good prediction of scaling behavior.

Doubling size, scales runtime by a constant for any polynomial time algorithm.

Not true for exponential algorithms. Could square runtime!

Asymptotic Analysis.

Used $O(n)$ for number of additions, rather than $n-2$.
Why?
61a, 61b..
Recursive fib has faster inner loop than iterative fib.
Does it matter?
 2^{694n} versus $O(n^2)$.
For 2^{694n} , doubling n , squares run time.
For $O(n^2)$, doubling n , multiplies run time by four.

Asymptotic Notation.

Ignore constant factors.
 $2n^2$ asymptotically same as $4n^2$
both are $O(n^2)$
 $4\log n$ asymptotically same as $100\log n$
both are $O(\log n)$
Ignore smaller order terms.
 $2n^2 + 100$ is $O(n^2)$
 $2n^2 + 1000\log n$ is $O(n^2)$
Upper bound.
 n^2 is $O(n^3)$.
 $\log n$ is $O(n)$.
Formally, for positive functions g, f from integers to reals,
 $g(n) = O(f(n))$, if there is a constant c where $g(n) \leq cf(n)$

More asymptotic notation.

Ω notation.
A "lower bound".
 $2n^2$ is $\Omega(n^2)$...and $\Omega(n)$...
Formally, for positive functions g, f from integers to reals,
 $g(n) = \Omega(f(n))$, if there is a constant c where $g(n) \geq cf(n)$
 $g(n) = \Theta(f(n))$ if $g(n) = O(f(n))$ and $g(n) = \Omega(f(n))$.

Arithmetic.

Addition: $O(n)$

1	0	0	0	0	1	1	1	1	
	1	2	3	4	5	6	7	8	9
+	9	2	1	2	3	7	6	9	1
	1	0	4	4	6	9	4	4	8
									0

Time: $O(n)$
Can we do better?
Need to write output ... optimal.

More Al Khwarizmi's: algorithms.

Addition: $O(n)$

Multiplication:

				1	2	3	4	5	6	7	8	9	
		×		9	2	1	2	3	7	6	9	1	
				1	2	3	4	5	6	7	8	9	
	9	2		2	2	2	2	2	2	2	2	1	
	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.

n

Time: $O(n^2)$

Multiplication

Multiplication: $O(n^2)$.

Is the best possible?

Every digit in x must multiply every digit in y at least once!
 $\Theta(n^2)$ such pairs.

Is this the best possible?

- (a) Yes.
- (b) No.

No.
What ?!?!
Really!
Later.

Exponentiation.

Compute x^y ?

If x and y are n -bit numbers,
tightest valid upper bound on the number of bits for x^y ...

- (a) $2n$
- (b) n^2
- (c) 2^n
- (d) $n2^n$

Number of bits is $\log_2 x^y = y \log x \leq 2^n \times n$.

Does it make sense to do this?

Seems better just to keep x and y around.

For Fun

You want to define function f, g that

f is applied to sets of size 40 of elements from 1, ..., 1000 and returns a set of size 41 of elements from 1 to 1000.

g is applied to sets of size 41 of elements from 1 to 1000 and returns a set of size 40 of elements from 1 to 1000.

And, for all sets of S of size 40 of elements from 1 to 1000, we have $g(f(S)) = S$.

...see you on Thu.